

PlantbreedAI Agent System Architecture

A full-stack blueprint of the R-powered AI intelligence for biometrics and genetics.



The Neural Infrastructure

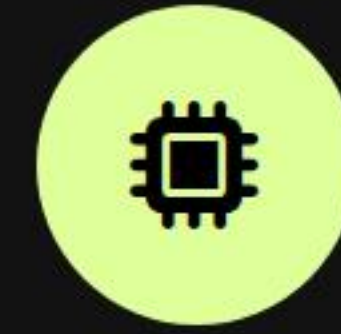
Mapping the path from User Interface to Statistical Execution.

Holistic Folder Hierarchy



Structural Core

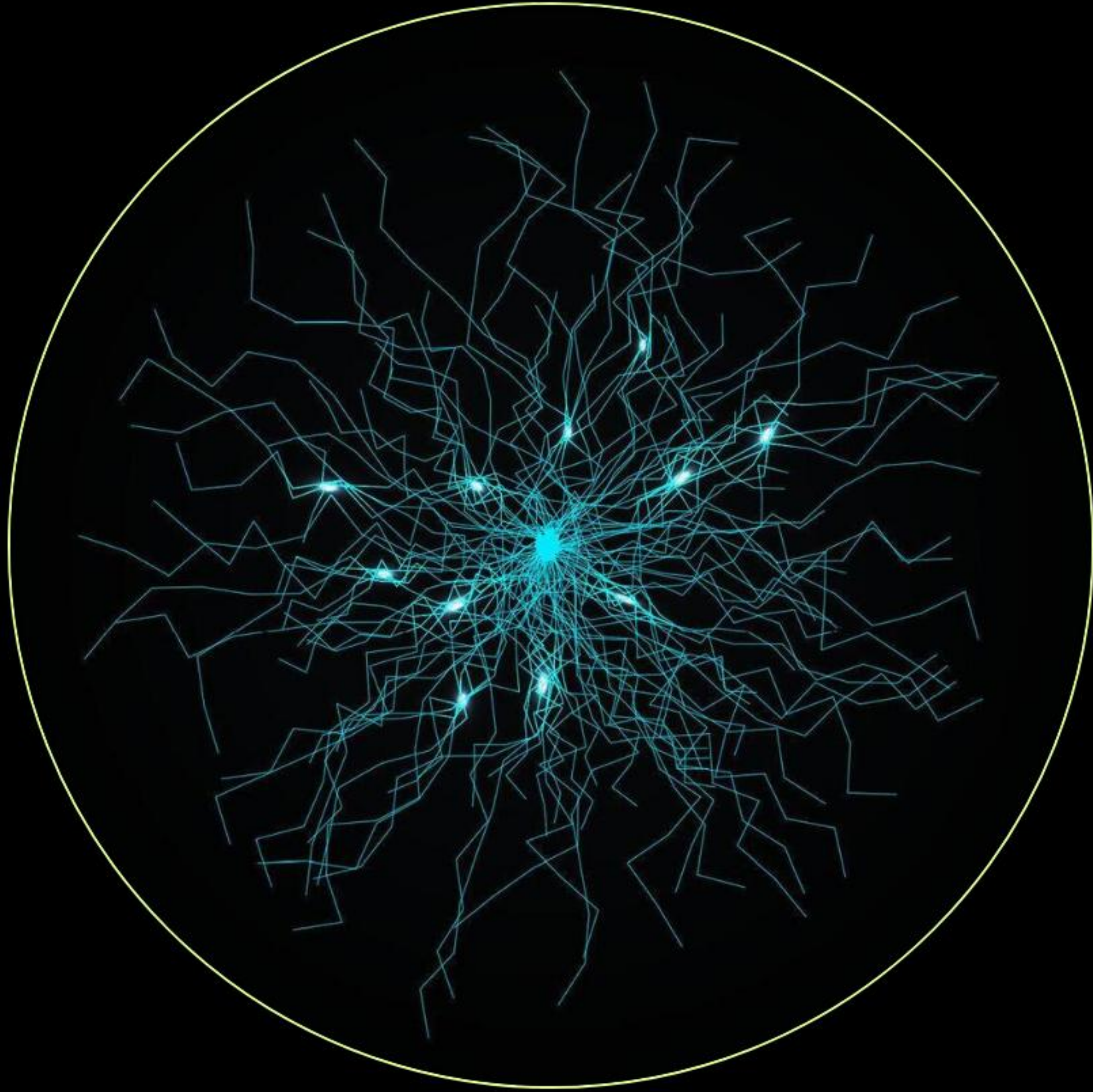
The root directory segments scientific logic (`core_logic`), AI intelligence (`agent_core`), and user interfacing (`frontend/api`), ensuring modular scalability.



Operational Flow

Standardized workflows transition data through the API bridge to the local LLM brain, triggering R-tools within a secure workspace environment.

The Neural Hub: `agent_core/`



Intelligent Orchestration

Orchestrator: The control tower managing Llama 3.2 tool-calling logic.

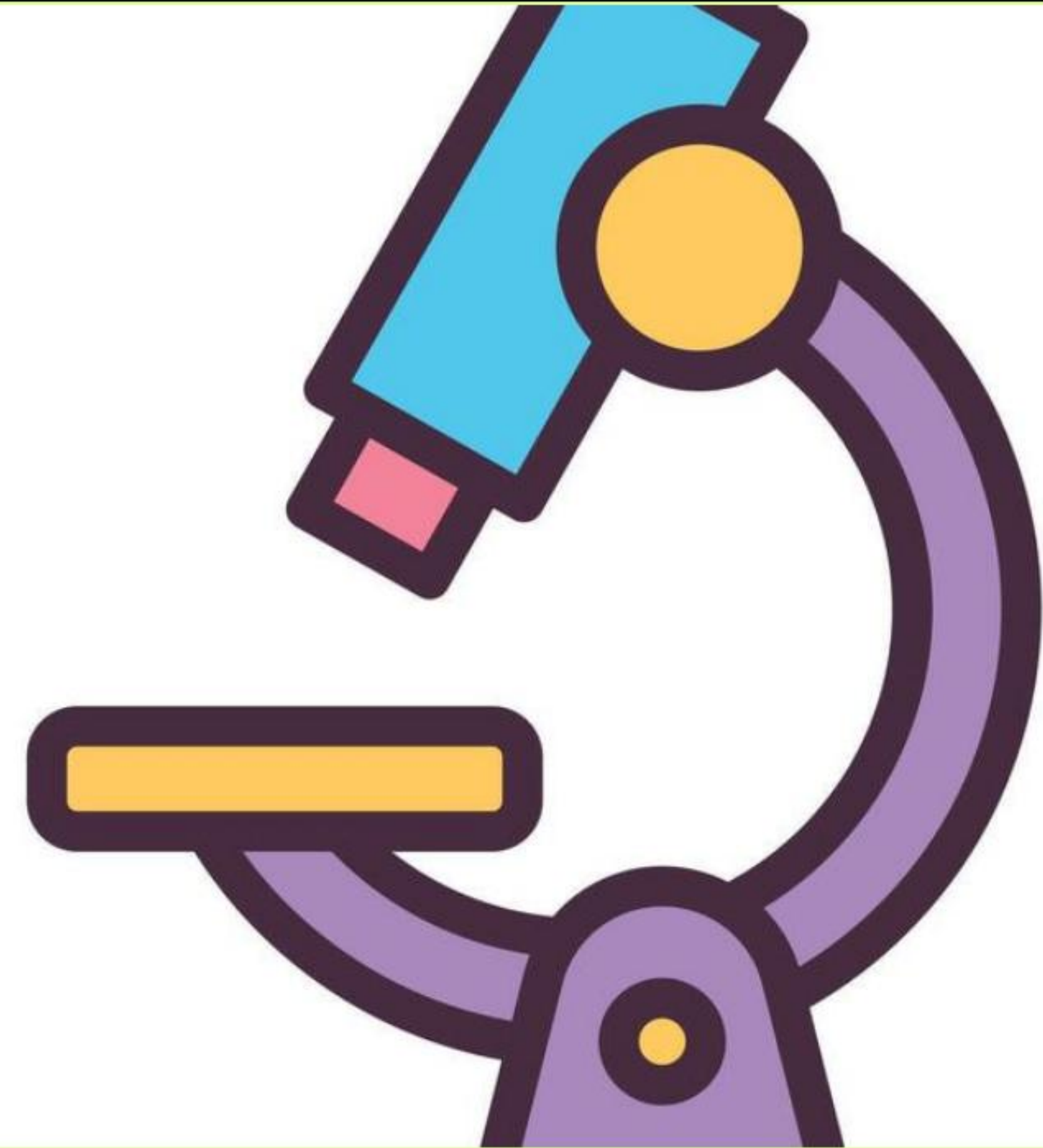
Prompts: Strict behavioral constraints preventing hallucination and ensuring scientific accuracy.

State: Persistent memory tracking chat history, active datasets, and last generated outputs.

FastAPI: The Communication Bridge

Seamless Integration

- **api/server.py**: The dedicated mouth and ears of the system.
- High-speed endpoint handling for asynchronous user requests.
- Initializes the AI Engine once at startup for maximum efficiency.
- Bridges the gap between the Python UI and the Local LLM Backend.



The Scientific Core

Genetic Logic Engine

Located in **core_logic/**, this layer houses the raw R libraries and Python scripts that define the mathematical reality of the agent.

It contains the complex biometrical functions, simulation models, and quantitative genetic parameters used to derive biological insights.



The Hands: **tools/** directory



R-Wrappers

Bilingual translators allowing Python/AI to trigger professional R scripts via rpy2 safely.



Tool Registry

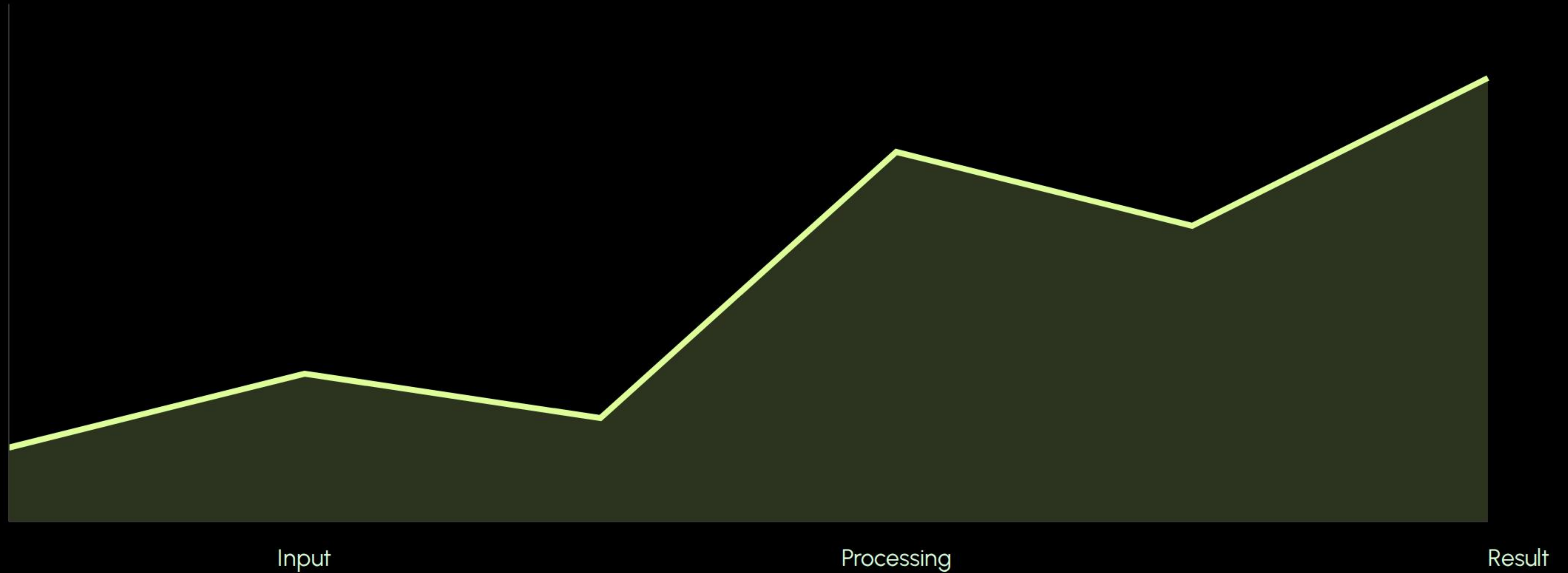
The master catalog of 40+ skills that the agent can perform, from AMMI to AUDPC.



Safe Dev-Off

Inbuilt safety handlers to prevent graphics engine crashes and null device errors.

Performance Execution Cycle



The "Thought-Action-Observation" loop optimizes processing time, scaling from simple simulation to complex MET analysis within seconds.

Workspace Data Logistics

44+

Scientific Tools

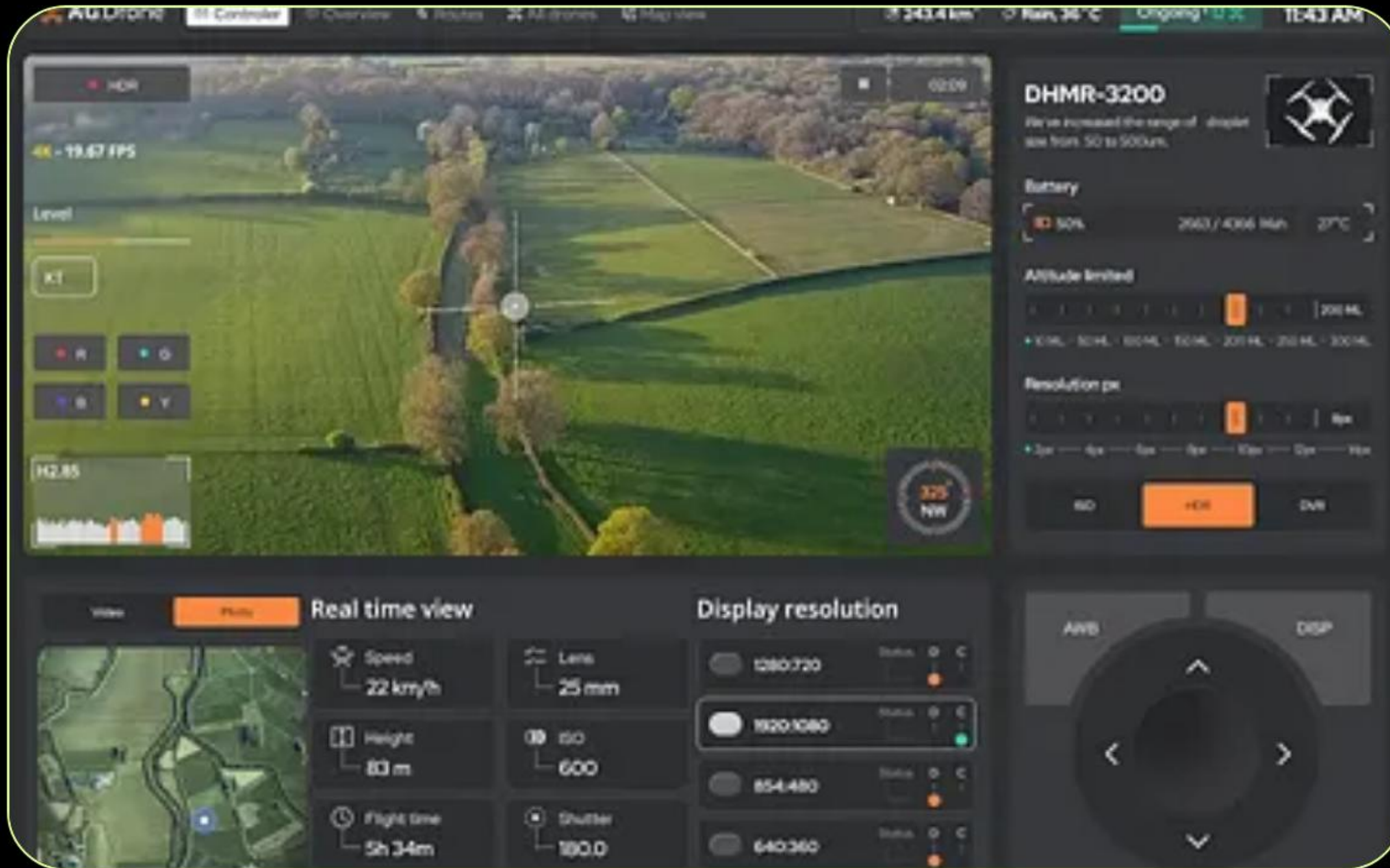
Data Segregation

Inputs: Secure staging for user CSV/XLSX files.

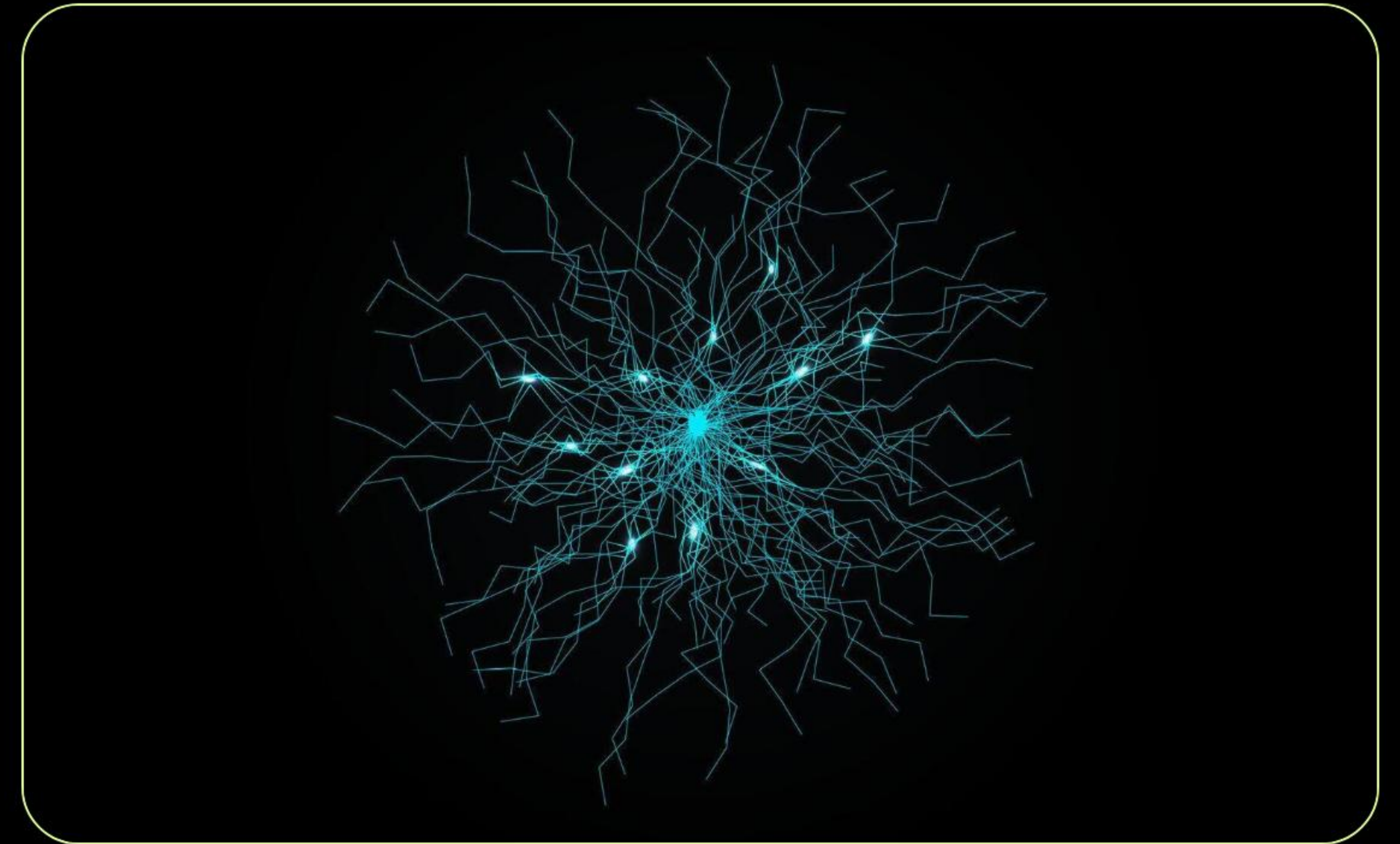
Outputs: Dynamic generation of PDFs, PNGs, and tables for real-time visualization.

Absolute Pathing: Prevents file hallucinations by forcing strict directory awareness.

User Experience: Dashboard Preview



Integrated UI with real-time chat and workspace visualization.



Deterministic AI logic ensures zero-hallucination statistical reporting.

Technical Stack Summary

- ✓ **Llama 3.2:** Local LLM providing high-level biometric reasoning.
- ✓ **R-Stats:** Industry-standard engine for quantitative genetics.
- ✓ **FastAPI:** Modern, high-performance API orchestration.
- ✓ **Streamlit:** Reactive frontend for real-time data visualization.

Questions?

Future-Proofing Plant Breeding with Agentic Intelligence.

Image Sources



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